Fiscal Year:	FY 2009	**Task Last Updated:**	FY 04/06/2010
PI Name:	Chaikin, Paul M. Ph.D.		
Project Title:	The Control and Dynamics of Hard Sphere Colloidal Dispersions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI moved to NYU (from Princeton U) in 2005 per A. Hollingsworth in PI's dept (7/2009). Changed email 3/30/2009 (chaikin@princeton.edu no longer valid).		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	01/14/2004	**End Date:**	01/13/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Meyer, William V	**Contact Phone:**	216-433-5011
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Flight Program:			
Flight Assignment:	note: grant renewed 1/2004 for 4 years/new nag#		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Russel, William (Princeton)		
Grant/Contract No.:	NNC04GA60G		
Performance Goal No.:	1.23		
Performance Goal Text:	Pursue ground-breaking basic research in fluid physics and transport phenomena to provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.		

Task Description:	This experiment consists of microscopy, microscopic and mesoscopic manipulation and control of colloidal samples to probe the essential features of the hard sphere disorder-order transition and the properties of the ordered solid phase that results. Many samples, in custom-designed microscope slide style cells that permit many different volume fractions and the application of external fields such as temperature gradients, periodic pattern imposition, and force application through the use of optical tweezers, will permit measurements from the disordered fluid phase, through the coexistence region, into the fully crystalline solid, and beyond into a glass-like phase. The dispersions will consist of colloidal hard spheres, in the 0.65 to 1.0 micron diameter range, with controlled polydispersity, consisting of poly (methyl methacrylate) (PMMA) in an index-matched fluid. Microscopy will allow for the observation of the position of individual colloidal particles and thus the determination of local crystalline or liquid structure, defect structure, observation of nucleation and growth of individual crystallites, the spontaneous formation of dendrites, the effect of periodic and other imposed patterns, field induced local melting and freezing, and the dynamics of liquid-crystal, liquid-glass and solid-glass interfaces. The applied fields will allow for controlled nucleation and growth, induced metastable phases, susceptibility, and rheology measurements. A simple design with many 50 microliter samples in a single sample holder on the microscope platen will allow interchange between the many samples to be studied. Static light scattering and confocal microscopy (replacing dynamic light scattering) will be done on the same samples. The static scattering image will be acquired with a CCD for Bragg scattering and structural studies. Confocal microscopy will allow the 3-d study of local affects. We expect to determine the volume fractions at which a fluid changes to a crystal, find the crystal structure of the equilibrium solid phase, contrast nucleation and growth in gravity and microgravity, especially in the region of the ground based glass transition (which we found does not occur in microgravity), quantitatively test the dendritic growth model, explore the role of polydispersity on the phase diagram and the structure of the different phases and explore the susceptibilities to external fields and the microrheology of the different phases. Characterization of crystal formation in the microgravity environment can lead to a greater understanding of how gravity affects many kinds of colloidal materials. Understanding these complex materials should enable new ways of forming ordered phases, such as those sought for photonic devices. The experiment employs a material that assumes multiple phases, which depend only on concentration. The simplest colloidal dispersion consists of hard spheres, which interact through hydrodynamic and Brownian forces, but feel no direct force before an infinite repulsion at contact. The hard sphere phase diagram, including the disorder-order transition, has been well established, but many-body interactions complicate the understanding of the kinetics of the transformations and the mechanical properties of disordered/metastable fluids and the ordered solid. Our experiments on these systems in microgravity, where the effects of sedimentation and convection are absent, have yielded several pleasant surprises. These included crystals (that were glasses on earth) and dendritic growth. In addition, Bragg images during nucleation and growth confirm some anticipated but previously unobserved features of the crystal growth process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Ground based support for our flight projects BCAT-4, BCAT-5 The Binary Colloidal Alloy Test (BCAT-4-Poly and BCAT-5-Poly) will be used to study the properties of concentrated systems comprised of small colloidal particles in an index matching fluid. In two samples, all but 1.0 and 0.2% of the spherical colloids are monodisperse ‘hard spheres’; the (minority) seed particles are about 11.5 times larger in diameter. These inclusions may induce crystallization by heterogeneous nucleation, an effect confounded by earth’s gravity. The theoretical prediction is that the use of the right size and concentration of seed particles can be used as a way to control the size of crystallites by reducing the free-energy barrier associated with crystal nucleation. The other BCAT samples consist of polydisperse and monodisperse colloidal glasses. In previous NASA microgravity experiments, we discovered that colloidal suspensions in the volume fraction range 0.58–0.64, which showed glassy behavior but did not crystallize in 1 g, readily formed crystals in microgravity. Although subsequent experiments with near-density-matched samples in normal gravity tended to confirm our results, the samples were far from ideal hard particles due to interactions inherent in the density matching process. Recent numerical simulations suggest that crystal nucleation in suspensions of colloidal spheres with broad size distributions should be suppressed; that is, vitrification at high supersaturations should yield colloidal glasses that are genuinely amorphous. We have extended our earlier microgravity experiments by studying the effects of polydispersity on crystallization in this volume fraction range. The control sample will be as monodispersed as possible, e.g., ~ 5%. Another class of colloidal particles prepared for BCAT experiments includes ellipsoids. We have found that depending on the ellipsoid shape, these particles can pack more densely than spheres in both the crystalline and random configurations. What is particularly interesting is that the difference in packing fraction between random and crystal is smaller for ellipsoids than for spheres; thus, the driving force for crystallization is reduced. Such behavior is reminiscent of the nonergodicity of glasses. In fact, following our work on granular ellipsoids, several groups have numerically tried to find a fluid-crystal transition from computer simulations. As yet there have been no reports of crystallization. While simulations and experiments with granular materials can provide information about packing they do not let us investigate the thermodynamic system the same way that experiments on colloidal systems will. We therefore propose a series of experiments on nonspherical, hard colloidal systems. Preliminary tests will be conducted using a BCAT cell filled with monodisperse hematite–silica (core-shell) ellipsoids. The PMMA colloidal spheres were prepared by dispersion polymerization using a poly(12-hydroxystearic acid)-g-poly(methyl methacrylate) stabilizer prepared in this lab. The 3.8 micron diameter PMMA particles were prepared by a seeded growth process, using PMMA particles prepared by the dispersion polymerization process. The polydisperse sample was prepared with a different batch of PHS-g-PMMA stabilizer that resulted in an unexpected polydispersity (approx. 11%) using twice the normal stabilizer addition (10% versus 5% based on the monomer charge). The scanning electron microscopy (SEM) pictures were taken at the Imaging and Analysis Center located at Princeton University. Small crystallites were observed in the polydisperse sample at effective volume fractions of 0.51 (just above the freezing transition) and 0.55 (just above the melting transition). SEM images were taken at the NYU Center for Soft Matter Research. The refractive index matching fluid utilized a mixture of 55% (w/w) tetrahydronaphthalene (tetralin) and 45% (w/w) decahydronaphthalene (decalin), a mixture of cis and trans isomers. The PMMA particles were thermally equilibrated with the solvent mixture to accelerate the particles swelling process. BCAT sample cells were filled Zin Technologies in Brookpark, OH. A safety report was submitted for the December 2007 NASA safety review: "Decalin hydroperoxide formation– Risk Assessment Report”.

	Hollow silica ellipsoids were prepared at NYU CSMR by dissolving the hematite cores with 20% hydrochloric acid. These particles have an aspect ratio of about 3:1 and will be refractive index matched with dimethylsulfoxide solvent. The synthesis followed the procedure developed by Sacanna et al. (2006). Hematite spindle-like particles stabilized by poly(vinylpyrrolidone) were prepared first. Using tetramethyl ammonium hydroxide, several shells of silica were grown via the polymerization of tetraethoxysilane. Several grams of material were produced. The ellipsoidal particles were intended to be flown on BCAT-5, but solvent compatibility issues prevented this. The dimethylsulfoxide (DMSO) solvent interacted with the Viton O-rings used for secondary confinement in the BCAT sample cell module. However, the Teflon-lined white rubber septa did seal the sample cells without leakage problems. The silica ellipsoids are planned for future microgravity experiments. Two batches of PMMA particles (8.1 g of ADH029 and 9 grams ADH030) were also prepared for Prof. Barbara Frisken (Simon Fraser University) and Dr. Art Bailey. These particles were used in their BCAT-5 sample preparation to study the competition between phase separation and crystallization. The latter batch was characterized with static and dynamic light scattering by Dr. Juan Sabin Fernandez, Simon Fraser University.
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